LangGraph + SQLite | Chatbot with Database Integration

1. Introduction

- **SQLite** is a lightweight relational database that stores data in a local file.
- In LangGraph, SQLite can be used to **persist chatbot state and conversation data**.
- This is useful when we want the chatbot to remember conversations even after the application restarts.

Basic idea:

User → Chatbot → LangGraph → SQLite

Instead of keeping state only in memory:

User → LangGraph → Memory lost after restart

we store it in SQLite:

User → LangGraph → SQLite → State/Conversation saved

2. Integrating Database

- A **checkpointer** is used in LangGraph to save the graph's state.
- SQLite can act as the storage system for this checkpointer.
- Each conversation can be associated with a **thread ID**.
- The thread ID allows the application to identify and retrieve a particular conversation.

Example:

Thread 1 → User's conversation A Thread 2 → User's conversation B Thread 3 → User's conversation C

When the user returns to Thread 1, LangGraph can retrieve its previously stored state from SQLite.

Why this is useful:

- Persistent conversation history
- Multiple conversations/users
- State survives application restart
- Easier conversation management
- Supports long-running applications

3. Frontend Code

- The frontend provides the interface through which the user interacts with the chatbot.
- It connects the UI to the LangGraph backend.
- When the user sends a message:
    1. Frontend sends the message and thread information.
    2. LangGraph processes the request.

3. The LLM generates a response.
4. Updated state is saved in SQLite.
5. The response is returned to the frontend.

Complete flow:

Frontend → LangGraph → LLM → SQLite (save state) → Frontend

The important concept is that **SQLite is not the LLM's memory itself**. It provides persistent storage that LangGraph can use to save and retrieve the graph's state.

Key Takeaway

LangGraph + SQLite = persistent chatbot state.

LangGraph manages the workflow and state, while **SQLite stores that state so the chatbot can continue conversations across sessions or application restarts.**